Physics 23: Section Worksheet

Div, Grad, Curl, and Electrostatic Fields

Detailed Step-by-Step Solution Key — Purcell 2.8 to 2.17

Spring 2026

Today's Toolkit — How the Four Laws Connect

$$\boxed{\nabla\cdot\vec{E} = \frac{\rho}{\varepsilon_0}} \qquad \boxed{\vec{E} = -\nabla\phi} \qquad \boxed{\nabla^2\phi = -\frac{\rho}{\varepsilon_0}} \qquad \boxed{\nabla\times\vec{E} = \vec{0}}$$

These are **not four separate facts**. They are four different lenses on the same field:

- $\nabla\cdot\vec{E} = \rho/\varepsilon_0$: **Divergence** counts net flux escaping a tiny volume — it detects local charge.
- $\vec{E} = -\nabla\phi$: **Gradient** tells you $\vec{E}$ points downhill in potential; ϕ increases opposite to $\vec{E}$.
- $\nabla^2\phi = -\rho/\varepsilon_0$: **Laplacian** measures whether ϕ curves up or down on average.
- $\nabla\times\vec{E} = \vec{0}$: **Curl** measures local circulation. Static $\vec{E}$ fields have none.

Problem 1

Two-Minute Warm-Up: Say It in Words

For each statement, determine: *true*, *false*, or *depends*. Give a one-phrase reason.

(a) If $\nabla\cdot\vec{E}(P) > 0$, then a tiny closed surface around P has net outward electric flux.

(b) If $\nabla\times\vec{E}(P) \neq \vec{0}$, then $\vec{E}$ cannot be a static electric field.

(c) If $\nabla\cdot\vec{E} = 0$ in a region, then $\vec{E} = \vec{0}$ in that region.

(d) If $\vec{E} = -\nabla\phi$, then moving around a closed loop changes ϕ by zero.

Step-by-Step Solution

(a) True.

Why? Divergence is defined as net outward flux per unit volume in the limit of a shrinking closed surface:

$$\nabla\cdot\vec{E}\big|_P = \lim_{V\to 0}\frac{1}{V}\oint_S \vec{E}\cdot d\vec{a}$$

If this limit is *positive*, the numerator $\oint_S \vec{E}\cdot d\vec{a}$ is also positive (for small but finite volume

V). Positive net outward flux means more field lines leave the surface than enter it. By Gauss's law, this signals a positive local charge density $\rho = \varepsilon_0 (\nabla \cdot \vec{E}) > 0$.

(b) True (for electrostatics).
Why? One of Maxwell's equations for electrostatics is $\nabla \times \vec{E} = \vec{0}$ everywhere in a static configuration. This is a *necessary* condition for $\vec{E}$ to be electrostatic. If we find any point where $\nabla \times \vec{E} \neq \vec{0}$, the field cannot be electrostatic.
Physically: the curl measures local circulation — the line integral of $\vec{E}$ around a tiny closed loop. A nonzero curl means $\vec{E}$ does net work on a test charge as it goes around a loop. Static electric fields never do this (they are conservative).

Common Mistake to Avoid

The statement says "*for electrostatics.*" If the problem were asking about a time-varying field (like an electromagnetic wave), $\nabla \times \vec{E} \neq \vec{0}$ is *allowed* by Faraday's law. But in this course we deal only with static fields.

(c) False.
Why? Zero divergence only tells you there are *no local sources or sinks* of flux — it says nothing about whether the field itself is zero.
Counterexamples:

- A uniform field $\vec{E} = E_0 \hat{x}$ has $\nabla \cdot \vec{E} = \partial E_0 / \partial x = 0$, yet $\vec{E} \neq \vec{0}$.
- The field outside a charged sphere ($r > R$) has $\nabla \cdot \vec{E} = 0$ everywhere in empty space, yet $\vec{E} \neq \vec{0}$ (it falls as $1/r^2$).

The divergence of $\vec{E}$ at a point only detects charge density *at that exact point.* A field produced by distant charges can pass through a charge-free point with zero local divergence but nonzero magnitude.

(d) True.
Why? The electrostatic potential ϕ is a single-valued scalar function of position. The line integral of $\vec{E} = -\nabla \phi$ along any path is:

$$\int_A^B \vec{E} \cdot d\vec{s} = -\int_A^B \nabla \phi \cdot d\vec{s} = -\big[\phi(B) - \phi(A)\big]$$

For a *closed* loop, the start and end points are the same: $A = B$. Therefore:

$$\oint_C \vec{E} \cdot d\vec{s} = -[\phi(A) - \phi(A)] = 0$$

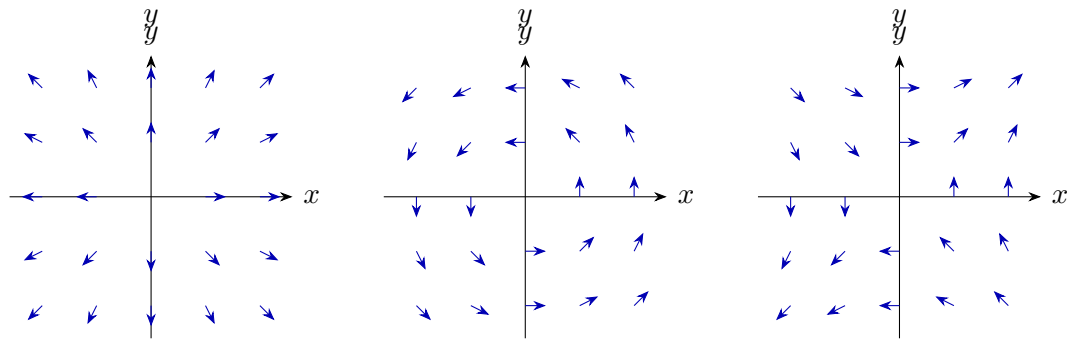
No matter what path you take, if you return to where you started, the total change in ϕ is zero. This is what "conservative field" means.

Problem 2

Fluxmeter vs. Curlmeter

Three vector fields in the xy-plane with $K > 0$, no z-component:

$$\vec{A} = K(x\hat{x} + y\hat{y}), \qquad \vec{B} = K(-y\hat{x} + x\hat{y}), \qquad \vec{C} = K(y\hat{x} + x\hat{y}).$$



A: $K(x\hat{x} + y\hat{y})$ **B:** $K(-y\hat{x} + x\hat{y})$ **C:** $K(y\hat{x} + x\hat{y})$

Step-by-Step Solution

Part (a): Visual inspection before calculating.

- **Field A** looks most like a *source of flux*: arrows point radially outward from the origin, spreading apart. Field lines start at the origin and radiate outward — this is what a positive charge source looks like.
- **Field B** looks most like *circulation*: arrows swirl counterclockwise around the origin without spreading. This looks like a whirlpool — characteristic of nonzero curl.
- **Field C** is the subtle case: arrows are tilted diagonally (pointing toward the line $y = x$ or $y = -x$). It looks neither purely radial nor purely swirling, but we must calculate carefully. *Visual appearance does not determine curl.*

Part (b): Calculate $\nabla \cdot \vec{E}$ and $(\nabla \times \vec{E})_z$ for each field.
Setup — the formulas to use: For a 2D field with no z-component and no z-dependence:

$$\nabla \cdot \vec{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \underbrace{\frac{\partial E_z}{\partial z}}_{=0}$$

$$(\nabla \times \vec{E})_z = \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}$$

(The x- and y-components of the curl are zero because $E_z = 0$ and there is no z-dependence.)

Field A: $E_x = Kx,\ \ E_y = Ky,\ \ E_z = 0.$
Step 1 — divergence:

$$\nabla\cdot\vec{A} = \frac{\partial(Kx)}{\partial x} + \frac{\partial(Ky)}{\partial y} = K\cdot\frac{\partial x}{\partial x} + K\cdot\frac{\partial y}{\partial y} = K(1) + K(1) = \boxed{2K}$$

Step 2 — curl:

$$(\nabla\times\vec{A})_z = \frac{\partial(Ky)}{\partial x} - \frac{\partial(Kx)}{\partial y} = K\underbrace{\frac{\partial y}{\partial x}}_{=0} - K\underbrace{\frac{\partial x}{\partial y}}_{=0} = \boxed{0}$$

Interpretation: $\nabla\cdot\vec{A} = 2K > 0$ means there is a net source of flux at every point (equivalently, a positive charge density $\rho = 2\varepsilon_0 K$). The zero curl means no circulation — consistent with the arrows pointing purely outward, never swirling.

Field B: $E_x = -Ky,\ \ E_y = Kx,\ \ E_z = 0.$
Step 1 — divergence:

$$\nabla\cdot\vec{B} = \frac{\partial(-Ky)}{\partial x} + \frac{\partial(Kx)}{\partial y} = -K\underbrace{\frac{\partial y}{\partial x}}_{=0} + K\underbrace{\frac{\partial x}{\partial y}}_{=0} = 0 + 0 = \boxed{0}$$

Step 2 — curl:

$$(\nabla\times\vec{B})_z = \frac{\partial(Kx)}{\partial x} - \frac{\partial(-Ky)}{\partial y} = K\underbrace{\frac{\partial x}{\partial x}}_{=1} - (-K)\underbrace{\frac{\partial y}{\partial y}}_{=1} = K - (-K) = \boxed{2K}$$

Interpretation: $\nabla\cdot\vec{B} = 0$ means field lines neither start nor end in the region — no local charge. The nonzero curl $(\nabla\times\vec{B})_z = 2K > 0$ confirms the visual impression: the field circulates counterclockwise everywhere, like a rigid rotation.

Field C: $E_x = Ky,\ \ E_y = Kx,\ \ E_z = 0.$
Step 1 — divergence:

$$\nabla\cdot\vec{C} = \frac{\partial(Ky)}{\partial x} + \frac{\partial(Kx)}{\partial y} = K\underbrace{\frac{\partial y}{\partial x}}_{=0} + K\underbrace{\frac{\partial x}{\partial y}}_{=0} = \boxed{0}$$

Step 2 — curl:

$$(\nabla\times\vec{C})_z = \frac{\partial(Kx)}{\partial x} - \frac{\partial(Ky)}{\partial y} = K\underbrace{\frac{\partial x}{\partial x}}_{=1} - K\underbrace{\frac{\partial y}{\partial y}}_{=1} = K - K = \boxed{0}$$

Interpretation: Both $\nabla\cdot\vec{C} = 0$ and $\nabla\times\vec{C} = \vec{0}$, even though the field looks complicated at first glance. The x-derivative of $E_y = Kx$ gives $+K$, and the y-derivative of $E_x = Ky$ also gives $+K$, and they cancel exactly in the curl formula.

Part (c): Which fields could be electrostatic in a charge-free region?
A valid electrostatic field in a charge-free region must satisfy **both**:

$$\nabla\cdot\vec{E} = 0 \quad \text{(no local charge)} \qquad \text{AND} \qquad \nabla\times\vec{E} = \vec{0} \quad \text{(no circulation)}$$

Field	$\nabla\cdot\vec{E}$	$(\nabla\times\vec{E})_z$	**Valid in charge-free region?**
A	$2K \neq 0$	0	**No** — nonzero divergence implies $\rho \neq 0$
B	0	$2K \neq 0$	**No** — nonzero curl forbidden for static $\vec{E}$
C	0	0	**Yes** — both conditions satisfied

Bonus: Find the potential ϕ for Field C.
If $\vec{E} = -\nabla\phi$, then $E_x = -\partial\phi/\partial x$ and $E_y = -\partial\phi/\partial y$.
Step 1: Match the x-component:

$$-\frac{\partial\phi}{\partial x} = Ky \implies \frac{\partial\phi}{\partial x} = -Ky \implies \phi = -Kxy + g(y)$$

where $g(y)$ is an unknown function of y (the "constant" of integration with respect to x).
Step 2: Match the y-component using what we found:

$$-\frac{\partial\phi}{\partial y} = Kx \implies \frac{\partial\phi}{\partial y} = -Kx$$

Differentiate $\phi = -Kxy + g(y)$ with respect to y:

$$\frac{\partial\phi}{\partial y} = -Kx + g'(y) = -Kx \implies g'(y) = 0 \implies g(y) = \text{const}$$

Result:

$$\boxed{\phi = -Kxy + \text{const}}$$

Common Mistake to Avoid

"Field C looks complicated, so it must have nonzero curl." This is one of the most important lessons of this problem. Curl is not about the visual complexity or curviness of arrows — it is a precise mathematical test ($\partial E_y/\partial x - \partial E_x/\partial y$). You must compute it. Never judge curl by appearance alone.

Problem 3

Same Sphere, Two Readings: Charge and Circulation

A uniformly charged solid sphere of radius R and volume charge density ρ:

$$\vec{E}(r) = \begin{cases} \dfrac{\rho}{3\varepsilon_0}\vec{r}, & r < R, \\ \dfrac{\rho R^3}{3\varepsilon_0 r^2}\hat{r}, & r > R. \end{cases} \qquad \text{where } \vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$$

Step-by-Step Solution

Part (a): Predict $\nabla\cdot\vec{E}$ inside and outside — physical argument.

Physical Intuition

The key insight: $\nabla\cdot\vec{E} = \rho/\varepsilon_0$ is a *local* equation. It says the divergence at a point equals the charge density *at that exact same point*. You do not need to know anything about the total charge or the field far away — only whether there is charge at the point you are examining.

Inside the sphere ($r < R$): The charge density is ρ (by assumption, it is uniform throughout the sphere). Applying Gauss's law in differential form directly:

$$\boxed{\nabla\cdot\vec{E} = \frac{\rho}{\varepsilon_0}} \qquad (r < R)$$

Outside the sphere ($r > R$): There is no charge in the empty space outside the sphere. The charge density is zero there:

$$\boxed{\nabla\cdot\vec{E} = 0} \qquad (r > R)$$

Part (b): Verify the inside result using Cartesian components.
Inside the sphere, write out the Cartesian components of $\vec{E} = \dfrac{\rho}{3\varepsilon_0}\vec{r}$:

$$E_x = \frac{\rho x}{3\varepsilon_0}, \qquad E_y = \frac{\rho y}{3\varepsilon_0}, \qquad E_z = \frac{\rho z}{3\varepsilon_0}$$

Now take each partial derivative one at a time:

$$\frac{\partial E_x}{\partial x} = \frac{\partial}{\partial x}\left(\frac{\rho x}{3\varepsilon_0}\right) = \frac{\rho}{3\varepsilon_0}\cdot\frac{\partial x}{\partial x} = \frac{\rho}{3\varepsilon_0}\cdot 1 = \frac{\rho}{3\varepsilon_0}$$

$$\frac{\partial E_y}{\partial y} = \frac{\partial}{\partial y}\left(\frac{\rho y}{3\varepsilon_0}\right) = \frac{\rho}{3\varepsilon_0}\cdot\frac{\partial y}{\partial y} = \frac{\rho}{3\varepsilon_0}\cdot 1 = \frac{\rho}{3\varepsilon_0}$$

$$\frac{\partial E_z}{\partial z} = \frac{\partial}{\partial z}\left(\frac{\rho z}{3\varepsilon_0}\right) = \frac{\rho}{3\varepsilon_0}\cdot\frac{\partial z}{\partial z} = \frac{\rho}{3\varepsilon_0}\cdot 1 = \frac{\rho}{3\varepsilon_0}$$

Adding them:

$$\nabla\cdot\vec{E} = \frac{\rho}{3\varepsilon_0} + \frac{\rho}{3\varepsilon_0} + \frac{\rho}{3\varepsilon_0} = \boxed{\frac{\rho}{\varepsilon_0}} \quad \checkmark$$

This agrees with our prediction from part (a) — a reassuring consistency check.

Part (c): Why is $\nabla\cdot\vec{E} = 0$ outside even though field lines spread?
The apparent paradox: outside the sphere, the field lines radiate outward and look like they are spreading (suggesting a source). But divergence is zero. Here is the resolution.

Physical Intuition

Key idea: Divergence measures whether flux is being *created* inside a tiny volume, not whether field lines spread geometrically in space.

Explicit check using flux: Consider two concentric spherical surfaces of radii r and $r+dr$ (both outside the sphere). The outward flux through each is:

$$\Phi(r) = E(r)\cdot 4\pi r^2 = \frac{\rho R^3}{3\varepsilon_0 r^2}\cdot 4\pi r^2 = \frac{4\pi\rho R^3}{3\varepsilon_0}$$

This is *independent of* r. The same total flux passes through every concentric sphere outside the charge distribution. Since no additional flux is created between neighboring spheres, the local divergence in that empty region is zero.
In other words: yes, the arrows spread out geometrically (the cross-section of the sphere grows as r^2), but simultaneously the magnitude decreases as $1/r^2$. The two effects cancel exactly, leaving zero net creation of flux per unit volume.

Part (d): What is $\nabla\times\vec{E}$ inside and outside? Give both reasons.

$$\boxed{\nabla\times\vec{E} = \vec{0} \quad \text{everywhere (inside and outside)}}$$

Mathematical reason: The field was found by applying Gauss's law and symmetry. It is the negative gradient of an electrostatic potential $\phi(r) = \text{const} - \int E\,dr$. Any field written as $\vec{E} = -\nabla\phi$ automatically has $\nabla\times\vec{E} = \vec{0}$ (this is the vector identity $\nabla\times(\nabla\phi) = \vec{0}$, which holds for any smooth ϕ).
Physical reason: This is a *static* electric field produced by a charge distribution at rest. By one of Maxwell's equations for statics, all electrostatic fields satisfy $\nabla\times\vec{E} = \vec{0}$. Equivalently, a static electric field is conservative — it does no net work on a test charge over any closed path. Zero net work per closed loop is exactly what zero curl expresses locally.

Problem 4

Can This Field Be Electrostatic? Can It Be Charge-Free?

For each field, determine: (1) Is it electrostatic? (2) Is the local charge density zero? (3) If electrostatic, find a potential ϕ.

$$\begin{aligned}
&\text{Field 1:} \quad K(y\hat{x} + x\hat{y}) \\
&\text{Field 2:} \quad K(-y\hat{x} + x\hat{y}) \\
&\text{Field 3:} \quad K(x\hat{x} + y\hat{y} + z\hat{z}) \\
&\text{Field 4:} \quad K(x\hat{x} + y\hat{y} - 2z\hat{z})
\end{aligned}$$

Step-by-Step Solution

The two tests — stated once, applied to all fields:

Question	Test	How to apply
Electrostatic?	$\nabla \times \vec{E} = \vec{0}$?	Compute $\partial E_y/\partial x - \partial E_x/\partial y$ (in 2D), or all three components (in 3D)
Charge-free?	$\nabla \cdot \vec{E} = 0$?	Compute $\partial E_x/\partial x + \partial E_y/\partial y + \partial E_z/\partial z$

Field 1: $\vec{E} = K(y\hat{x} + x\hat{y})$, so $E_x = Ky$, $E_y = Kx$, $E_z = 0$.
Divergence:

$$\nabla \cdot \vec{E}_1 = \frac{\partial(Ky)}{\partial x} + \frac{\partial(Kx)}{\partial y} + 0 = 0 + 0 = 0 \;\Rightarrow\; \rho = \varepsilon_0 \cdot 0 = 0. \quad \text{Charge-free: } \textbf{YES}.$$

Curl (z-component only, since field is 2D):

$$(\nabla \times \vec{E}_1)_z = \frac{\partial(Kx)}{\partial x} - \frac{\partial(Ky)}{\partial y} = K(1) - K(1) = 0. \quad \text{Electrostatic: } \textbf{YES}.$$

Find potential ϕ: We need $E_x = -\partial\phi/\partial x$ and $E_y = -\partial\phi/\partial y$.
Step 1 — integrate $\partial\phi/\partial x = -Ky$ with respect to x:

$$\phi = -Kxy + g(y)$$

Step 2 — use the y-condition: differentiate with respect to y:

$$\frac{\partial\phi}{\partial y} = -Kx + g'(y) \stackrel{!}{=} -Kx \;\Rightarrow\; g'(y) = 0 \;\Rightarrow\; g = \text{const}$$

$$\boxed{\phi_1 = -Kxy + C}$$

Quick check: $-\partial\phi/\partial x = Ky = E_x$ ✓ and $-\partial\phi/\partial y = Kx = E_y$ ✓.

Field 2: $\vec{E} = K(-y\hat{x} + x\hat{y})$, so $E_x = -Ky$, $E_y = Kx$, $E_z = 0$.
Divergence:

$$\nabla\cdot\vec{E}_2 = \frac{\partial(-Ky)}{\partial x} + \frac{\partial(Kx)}{\partial y} = 0 + 0 = 0. \quad \text{Charge-free: } \textbf{YES}.$$

Curl:

$$(\nabla\times\vec{E}_2)_z = \frac{\partial(Kx)}{\partial x} - \frac{\partial(-Ky)}{\partial y} = K(1) - (-K)(1) = K + K = 2K \neq 0. \quad \text{Electrostatic: } \textbf{NO}.$$

Since the curl is nonzero, this field *cannot* be written as $-\nabla\phi$, and no potential exists. No further calculation needed.

Field 3: $\vec{E} = K(x\hat{x} + y\hat{y} + z\hat{z})$, so $E_x = Kx$, $E_y = Ky$, $E_z = Kz$.
Divergence:

$$\nabla\cdot\vec{E}_3 = \frac{\partial(Kx)}{\partial x} + \frac{\partial(Ky)}{\partial y} + \frac{\partial(Kz)}{\partial z} = K + K + K = 3K. \quad \text{Charge-free: } \textbf{NO} \ (\rho = 3\varepsilon_0 K).$$

Curl (all three components):

$$\begin{aligned}
(\nabla\times\vec{E}_3)_x &= \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = \frac{\partial(Kz)}{\partial y} - \frac{\partial(Ky)}{\partial z} = 0 - 0 = 0 \\
(\nabla\times\vec{E}_3)_y &= \frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} = \frac{\partial(Kx)}{\partial z} - \frac{\partial(Kz)}{\partial x} = 0 - 0 = 0 \\
(\nabla\times\vec{E}_3)_z &= \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = \frac{\partial(Ky)}{\partial x} - \frac{\partial(Kx)}{\partial y} = 0 - 0 = 0
\end{aligned}$$

$\nabla\times\vec{E}_3 = \vec{0}$, so: Electrostatic: **YES** (but not charge-free).
Find potential ϕ:
Step 1 — from $E_x = Kx$: $\partial\phi/\partial x = -Kx \Rightarrow \phi = -\frac{K}{2}x^2 + f(y,z)$
Step 2 — from $E_y = Ky$: $\partial\phi/\partial y = -Ky$. Differentiate ϕ found above:

$$\frac{\partial\phi}{\partial y} = f_y(y,z) = -Ky \Rightarrow f = -\frac{K}{2}y^2 + g(z)$$

Step 3 — from $E_z = Kz$: $\partial\phi/\partial z = -Kz$. Differentiate:

$$\frac{\partial\phi}{\partial z} = g'(z) = -Kz \Rightarrow g = -\frac{K}{2}z^2 + C$$

Result:

$$\boxed{\phi_3 = -\frac{K}{2}(x^2 + y^2 + z^2) + C}$$

Field 4: $\vec{E} = K(x\hat{x} + y\hat{y} - 2z\hat{z})$, so $E_x = Kx$, $E_y = Ky$, $E_z = -2Kz$.

Divergence:

$$\nabla\cdot\vec{E}_4 = \frac{\partial(Kx)}{\partial x} + \frac{\partial(Ky)}{\partial y} + \frac{\partial(-2Kz)}{\partial z} = K + K + (-2K) = \boxed{0}. \quad \text{Charge-free: } \textbf{YES}.$$

The $+K + K$ from the x and y components is exactly cancelled by the $-2K$ from the z component. This is not a coincidence — see part (c).
Curl: Same structure as Field 3 but with an extra z-component. Each cross-derivative of distinct components (E_z w.r.t. y, etc.) is zero because each component only depends on its own variable. Therefore $\nabla\times\vec{E}_4 = \vec{0}$. Electrostatic: **YES**.
Find potential ϕ:
Step 1: $\partial\phi/\partial x = -Kx \Rightarrow \phi = -\frac{K}{2}x^2 + f(y,z)$
Step 2: $f_y = -Ky \Rightarrow f = -\frac{K}{2}y^2 + g(z)$
Step 3: $g'(z) = \partial\phi/\partial z = 2Kz \Rightarrow g = Kz^2 + C$
Result:

$$\boxed{\phi_4 = -\frac{K}{2}(x^2 + y^2 - 2z^2) + C}$$

Summary table:

Field	$\vec{E}$	**Electrostatic?**	$\rho = 0$**?**	**One possible** ϕ
1	$K(y\hat{x} + x\hat{y})$	Yes	Yes	$-Kxy$
2	$K(-y\hat{x} + x\hat{y})$	No	Yes	none
3	$K(x\hat{x} + y\hat{y} + z\hat{z})$	Yes	No	$-\frac{K}{2}(x^2 + y^2 + z^2)$
4	$K(x\hat{x} + y\hat{y} - 2z\hat{z})$	Yes	Yes	$-\frac{K}{2}(x^2 + y^2 - 2z^2)$

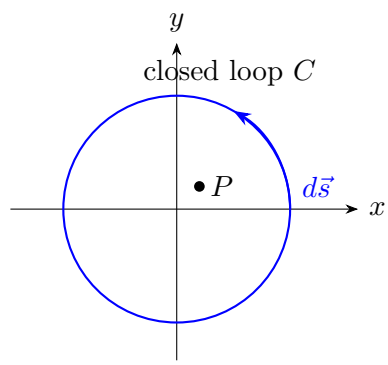
Part (c): What Field 4 teaches about Laplace's equation.
For Field 4, $\nabla\cdot\vec{E} = 0$ and the field is electrostatic, so the potential satisfies $\nabla^2\phi = 0$ (Laplace's equation). But $\vec{E} \neq \vec{0}$ and ϕ is not constant. What does this mean?
$\nabla^2\phi = 0$ says the potential is **harmonic** — it has zero net curvature (no preference for curving up or down on average). Any upward curvature in one direction is exactly offset by downward curvature in another direction. Concretely, for $\phi_4 = -\frac{K}{2}(x^2 + y^2 - 2z^2)$:

$$\frac{\partial^2\phi}{\partial x^2} = -K, \quad \frac{\partial^2\phi}{\partial y^2} = -K, \quad \frac{\partial^2\phi}{\partial z^2} = +2K$$

These sum to zero. ϕ_4 curves *downward* in x and y (like a bowl) but *upward* in z (like a hill) — a **saddle-shaped potential**. This is the fingerprint of a charge-free electrostatic region: the potential always looks like a saddle, never like a pure bowl or pure hill.

Problem 5

Why $\nabla \times (\nabla\phi) = \vec{0}$ Matters



Step-by-Step Solution

Part (a): Why $\oint_C \vec{E} \cdot d\vec{s} = 0$ for every closed loop.
Given: $\vec{E} = -\nabla\phi$.
Step 1: Write the line integral:

$$\oint_C \vec{E} \cdot d\vec{s} = -\oint_C \nabla\phi \cdot d\vec{s}$$

Step 2: Recall what the line integral of a gradient means. For *any* two points A and B, the fundamental theorem of calculus for gradients says:

$$\int_A^B \nabla\phi \cdot d\vec{s} = \phi(B) - \phi(A)$$

This is path-independent — it only depends on the values of ϕ at the endpoints.
Step 3: Apply to a closed loop. A closed loop starts and ends at the *same* point, call it $A = B$:

$$\oint_C \nabla\phi \cdot d\vec{s} = \phi(A) - \phi(A) = 0$$

Therefore:

$$\boxed{\oint_C \vec{E} \cdot d\vec{s} = 0}$$

Physical meaning: The electric potential ϕ is a single-valued function. No matter what path a test charge takes, when it returns to its starting point, ϕ returns to its original value. The work done by $\vec{E}$ on a round trip is zero — the electrostatic force is conservative.

Part (b): Using Stokes' theorem to conclude $\nabla \times \vec{E} = \vec{0}$.

Stokes' theorem relates a line integral around a closed loop C to a surface integral over any surface S bounded by C:

$$\oint_C \vec{E} \cdot d\vec{s} = \iint_S (\nabla \times \vec{E}) \cdot d\vec{a}$$

From part (a), the left side is zero for *every* closed loop C in an electrostatic field.
Now suppose $\nabla \times \vec{E} \neq \vec{0}$ at some point P. Then we could choose a tiny loop C encircling P and a tiny flat surface S through P. The integral $\iint_S (\nabla \times \vec{E}) \cdot d\vec{a}$ would be nonzero (because $(\nabla \times \vec{E})(P) \neq 0$ and S is tiny, so the integral is approximately $(\nabla \times \vec{E})(P) \cdot \hat{n}\, \Delta A \neq 0$). This contradicts the left side being zero.
The only way both sides can be zero for *every* possible loop C (and hence every possible surface S through every point) is:

$$\boxed{\nabla \times \vec{E} = \vec{0} \text{ everywhere}}$$

Part (c): Counterexample — "zero curl" does not mean "straight field lines."

Common Mistake to Avoid

Student misconception: "If $\nabla \times \vec{E} = 0$, then field lines cannot curve."
This confuses the *curvature of field lines* (a geometric property of how the arrows point) with *curl* (a measurement of circulation around an infinitesimal loop). These are completely different things.

Counterexample 1 — point charge: The field of a point charge q is:

$$\vec{E} = \frac{q}{4\pi\varepsilon_0 r^2}\hat{r}$$

The field lines radiate outward — they are *straight* (radial). But consider the space away from the origin: $\nabla \times \vec{E} = \vec{0}$ everywhere (away from the source). Here field lines are straight *and* curl is zero — consistent.
Counterexample 2 — electric dipole: The field lines of an electric dipole curve from the $+q$ end, arch through space, and re-enter at the $-q$ end. The lines are clearly *curved*. Yet for a static dipole, $\nabla \times \vec{E} = \vec{0}$ everywhere (away from the point charges themselves). Curved field lines, zero curl.
The key: Curl is about the *line integral around a loop*, not about whether individual field lines bend. Even when field lines curve, there can be no net circulation around any tiny loop.

Part (d): Counterexample — "zero divergence" does not mean "no field."

Common Mistake to Avoid

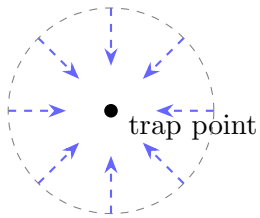
Student misconception: "If $\nabla \cdot \vec{E} = 0$, then there is no electric field."
This confuses *no local source of flux* with *no flux at all*. Divergence only detects whether flux is being created or destroyed *locally*. It says nothing about whether flux produced by distant charges is passing through the point.

Counterexample 1 — uniform field: $\vec{E} = E_0\hat{x}$ has $\nabla \cdot \vec{E} = \partial E_0/\partial x = 0$ everywhere, yet $|\vec{E}| = E_0 \neq 0$.
Counterexample 2 — outside a charged sphere: In the empty region $r > R$ outside a uniformly charged sphere, $\nabla \cdot \vec{E} = 0$ (no local charge). But the field $\vec{E} = (\rho R^3/3\varepsilon_0 r^2)\hat{r}$ is clearly nonzero. The field was created by charges inside the sphere, not by local sources in the empty region.
The key: Divergence is a *local* operator. It tells you about charge density at the single point where you evaluate it, not about fields arriving from elsewhere.

Problem 6

Earnshaw's Theorem as a Local Diagnostic

Can you trap a positive test charge at the origin using only static electric fields in empty space?



"restoring" field near a stable trap

Step-by-Step Solution

Part (a): What does the field look like near a stable trap?
For a *positive* test charge, the force is $\vec{F} = q\vec{E}$ (with $q > 0$), so force and field point in the same direction.
For a *stable* trap, a small displacement in *any* direction must produce a restoring force — a force pointing back *toward the origin*. This means the force must point *inward* everywhere on a small sphere surrounding the trap point.
Since $\vec{F} = q\vec{E}$ and $q > 0$, the field $\vec{E}$ must also point *inward* on that tiny sphere.
What does this imply about the flux? Inward-pointing field means *negative* outward flux:

$$\oint_S \vec{E} \cdot d\vec{a} < 0 \quad \text{(net inward flux)}$$

Part (b): Why is this impossible in empty space?
By Gauss's law:

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

From part (a), a stable trap requires the left side to be negative: $Q_{\text{enc}} < 0$.
But the trap point is supposed to be in *empty space*. Empty space means no charge inside the small sphere: $Q_{\text{enc}} = 0$ for a sufficiently small sphere.
Contradiction: the trap requires $Q_{\text{enc}} < 0$, but empty space gives $Q_{\text{enc}} = 0$.
Therefore, **no stable electrostatic trap exists in empty space.** This is Earnshaw's theorem.

Part (c): Proposed trap potential $\phi = \phi_0 + \frac{\alpha}{2}(x^2 + y^2 + z^2)$, $\alpha > 0$.
This has a minimum at the origin (since $x^2 + y^2 + z^2 \geq 0$, the minimum is at $x = y = z = 0$).

Step 1 — compute $\nabla^2\phi$:

$$\frac{\partial^2\phi}{\partial x^2} = \frac{\partial^2}{\partial x^2}\left(\phi_0 + \frac{\alpha}{2}x^2 + \frac{\alpha}{2}y^2 + \frac{\alpha}{2}z^2\right) = \frac{\alpha}{2}\cdot\frac{\partial^2 x^2}{\partial x^2} = \frac{\alpha}{2}\cdot 2 = \alpha$$
$$\frac{\partial^2\phi}{\partial y^2} = \alpha \quad \text{(same calculation)}$$
$$\frac{\partial^2\phi}{\partial z^2} = \alpha \quad \text{(same calculation)}$$

Adding:

$$\nabla^2\phi = \alpha + \alpha + \alpha = 3\alpha$$

Step 2 — apply Poisson's equation:

$$\nabla^2\phi = -\frac{\rho}{\varepsilon_0} \implies 3\alpha = -\frac{\rho}{\varepsilon_0} \implies \boxed{\rho = -3\varepsilon_0\alpha}$$

Interpretation: Since $\alpha > 0$, we need $\rho = -3\varepsilon_0\alpha < 0$ — a *negative* charge density at the origin. The potential minimum only exists because there is negative charge there drawing the positive test charge in. Without that negative charge (i.e., in empty space where $\rho = 0$), the Laplacian of ϕ cannot be positive, so ϕ cannot have a minimum.

Part (d): Two equal positive charges — equilibrium but not a trap.
Place charges $+Q$ at $x = +d$ and $x = -d$. At the midpoint $x = 0$, the forces from the two charges cancel by symmetry, so $\vec{E} = \vec{0}$ there. This is an *equilibrium*, but is it *stable*?
Step 1 — displacement along the axis (say $+\hat{x}$): If the test charge moves slightly toward the right charge ($+Q$ at $+d$), the right charge exerts a stronger repulsion (closer) than the left charge. Net force: pushes *back toward the midpoint*. This direction is *restoring*. ✓
Step 2 — displacement perpendicular to the axis (say $+\hat{y}$): If the test charge moves slightly upward, both $+Q$ charges now pull it (via the y-components of $\vec{r}_1$ and $\vec{r}_2$) slightly downward and outward. Wait — let's think more carefully. Both charges are at $(\pm d, 0)$. A test charge displaced to $(0, \epsilon)$ is equidistant from both charges. Each repels it in the direction from that charge to the test charge, which has both an inward x-component (cancels by symmetry) and an *upward* y-component. Net force: upward, *away from the equilibrium*. This direction is *anti-restoring*. ×
The equilibrium is stable along the axis but unstable perpendicular to it. It is a **saddle point** of the potential energy, not a minimum. Earnshaw's theorem is not violated: a saddle point equilibrium is not a stable trap.

Problem 7

Optional Extension: Stokes' Theorem as a Shortcut

Let $\vec{E} = K(-y\hat{x} + x\hat{y})$ with $K > 0$. Let C be a circle of radius a in the xy-plane, centered at the origin, oriented counterclockwise as viewed from $+z$.

Step-by-Step Solution

Part (a): Compute $\oint_C \vec{E} \cdot d\vec{s}$ using curl — no line integral required.

Step 1 — find $\nabla \times \vec{E}$.

We already computed this in Problem 2 (Field B):

$$(\nabla \times \vec{E})_x = \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = 0 - \frac{\partial (Kx)}{\partial z} = 0$$
$$(\nabla \times \vec{E})_y = \frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} = \frac{\partial (-Ky)}{\partial z} - 0 = 0$$
$$(\nabla \times \vec{E})_z = \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = \frac{\partial (Kx)}{\partial x} - \frac{\partial (-Ky)}{\partial y} = K - (-K) = 2K$$

So $\nabla \times \vec{E} = 2K\hat{z}$ — a uniform vector field pointing in the $+z$ direction everywhere.

Step 2 — apply Stokes' theorem.

Let S be the flat disk of radius a in the xy-plane bounded by the loop C. The outward normal to S (consistent with the counterclockwise orientation of C by the right-hand rule) is $\hat{n} = +\hat{z}$.

$$\oint_C \vec{E} \cdot d\vec{s} = \iint_S (\nabla \times \vec{E}) \cdot d\vec{a} = \iint_S (2K\hat{z}) \cdot (\hat{z}\, dA) = 2K \iint_S dA$$

Step 3 — evaluate the area integral.

The area of the disk of radius a is πa^2:

$$\oint_C \vec{E} \cdot d\vec{s} = 2K \cdot \pi a^2 = \boxed{2\pi K a^2}$$

Why is this a "shortcut"? The direct line integral requires parametrizing the circle as $x = a\cos t$, $y = a\sin t$, computing $E_x\, dx + E_y\, dy$, and integrating from $t = 0$ to 2π. Stokes' theorem reduced this to multiplying the uniform curl by the area of the disk — much simpler.

Part (b): Why does this nonzero result rule out the field as electrostatic?

For a static electric field, $\nabla \times \vec{E} = \vec{0}$ everywhere. By Stokes' theorem, this forces:

$$\oint_C \vec{E} \cdot d\vec{s} = \iint_S \underbrace{(\nabla \times \vec{E})}_{\vec{0}} \cdot d\vec{a} = 0$$

for *every* closed loop.

But we found $\oint_C \vec{E} \cdot d\vec{s} = 2\pi K a^2 \neq 0$. This is a nonzero line integral around a closed loop. If we were to move a positive test charge around this circle, the field $\vec{E}$ would do positive net work on it.

In contrast, *a conservative (electrostatic) field can never do net work around a closed loop* — that is precisely what path-independence means. A loop integral of $2\pi K a^2$ directly proves the field is non-conservative and therefore *cannot* be a static electric field.

Physical Intuition

Summary of the connection: All three statements below say exactly the same thing, and Field B violates all three.

$$\begin{aligned} \text{Electrostatic} \;&\Leftrightarrow\; \nabla \times \vec{E} = \vec{0} \\ &\Leftrightarrow\; \oint_C \vec{E} \cdot d\vec{s} = 0 \;\; \text{for every closed loop } C \\ &\Leftrightarrow\; \vec{E} = -\nabla\phi \;\; \text{for some scalar potential } \phi \end{aligned}$$